Additional Samples FOR CHAPTEr 4

CSP 650

Project

Chapter 4: Result / Findings/ Analysis



suhailayani@uitm.edu.my

Introduction : Findings

1. Reminding the reader of what your solution mapped to the problem.
2. A brief description of how you intend approaching the write up of the results.
3. Placing the result in context (map with title and objectives).
4. Letting the reader know where they can find the research instruments (i.e. the Appendix)



suhailayani@uitm.edu.my

Structure

- The results section should simply **state the findings, without bias or interpretation, and arranged in a logical sequence**. The results section should always be written in the past tense.

1. Present a synopsis of the results followed by an explanation of key findings. This approach can be used to highlight important findings of xxxxx (your title/ your main use case/ your main objectives)
2. Present a result and then explain it, before presenting the next result then explaining it, and so on, then end with an overall synopsis.



Sample 1 : Data Mining results



suhailayani@uitm.e

CHAPTER 4

FINDINGS AND ANALYSIS

This chapter elaborates more on the results of the data in conjunction with the methodology discussed in the Chapter 3. This chapter also describe about the findings and analysis of the data.

1.1 Phase 1: Gathering data and analyzed requirement

The first phase of this project is achieved by gathering the data from the stakeholder. This is the main process in the KDD methodology where it acts as the first phase to determine the main problem that is encountered by the stakeholder. The data given is in the form of tabulated data. To analyze the data there are a few methods that need to be done. This phase is one of the phase that need to be done in order to achieve the first objective which is to gather and analyze the requirement of the water quality data in Chini Lake.

- **Gather the requirement and data from the stakeholder**

At this activity, all the data has been collected and identified. The data are temperature, pH value, turbidity, TDS, conductivity, ODO, rainfall, and water level. All the data is considered as a raw data. The data were given through email since the data is huge because it contains from the year 2011 until 2014.

	Time	Temperature(°C)	Cond(uS/cm)	TDS(mg/l)	pH(unit)	Turbidity(NTU)	ODO(mg/l)
2012/01/01 12:00 AM	2012/01/01 12:00 AM	26.07	25	0.016	3.6	1014.70001	0.29
2012/01/01 08:00 AM	2012/01/01 08:00 AM	25.9	26	0.017	3.77	1005.90002	0.14
2012/01/01 04:00 PM	2012/01/01 04:00 PM	26.26	26	0.016	3.79	989.59998	0.14
2012/01/02 12:00 AM	2012/01/02 12:00 AM	26.93	28	0.017	3.76	929.29999	0.22
2012/01/02 08:00 AM	2012/01/02 08:00 AM	27.03	26	0.016	3.72	909.79999	0.15
2012/01/02 04:00 PM	2012/01/02 04:00 PM	26.93	25	0.016	3.63	916.40002	0.34
2012/01/03 12:00 AM	2012/01/03 12:00 AM	26.91	26	0.016	3.58	917.59998	0.59
2012/01/03 04:00 PM	2012/01/03 08:00 AM	26.78	27	0.017	3.55	908.79999	0.49
2012/01/03 08:00 AM	2012/01/03 04:00 PM	26.72	25	0.016	3.56	900.79999	0.26
2012/01/04 12:00 AM	2012/01/04 12:00 AM	26.7	26	0.016	3.63	884.5	0.37
2012/01/04 08:00 AM	2012/01/04 08:00 AM	26.54	27	0.017	3.65	889.5	0.29
2012/01/04 04:00 PM	2012/01/04 04:00 PM	26.61	26	0.016	3.7	841.20001	0.17
2012/01/05 12:00 AM	2012/01/05 12:00 AM	26.65	26	0.017	3.74	875.70001	0.15
2012/01/05 08:00 AM	2012/01/05 08:00 AM	26.48	32	0.02	3.68	861.59998	0.33
2012/01/05 04:00 PM	2012/01/05 04:00 PM	26.56	27	0.017	3.66	802.70001	0.14
2012/01/06 12:00 AM	2012/01/06 04:00 PM	26.68	30	0.019	3.66	781.79999	0.1
	2012/01/06 12:00 AM						
	2012/01/06 04:00 AM						
	2012/01/07 12:00 AM	26.73	31	0.02	3.66	735.5	0.15
	2012/01/07 08:00 AM	26.54	34	0.021	3.66	690.90002	0.16
2012/01/07 12:00 AM	2012/01/07 04:00 PM	26.72	33	0.021	3.71	668.70001	0.07
2012/01/07 08:00 AM	2012/01/08 12:00 AM						
2012/01/07 04:00 PM	2012/01/08 08:00 AM						
	2012/01/08 04:00 PM						

Figure 4.1 Raw data of water quality

Sample 1 : Data Mining results



1.2 Phase 2: Data Selection

Phase 2 is the process of selecting suitable data to be analyzed further in this project. This include reviewing the data and select the targeted data. This phase 2 also will contribute in achieving the first objective.

- **Review all data and select targeted data**

Selection of the right and acceptable data is important in order to acquire a suitable data that will be used for the project because not all data that was given is suitable be used in the project. The data need to be reviewed and analyze in order to acquire only the important and the right data to be used in the development of the project. There a few data that is selected in this project which is Chini Lake data. Meanwhile, there are also data that is filtered and not needed for the project the data from other lakes.

date	temp	cond	TDS	pH	turb	DO
1/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
2/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
3/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
4/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
5/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
6/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
7/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
8/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
9/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
10/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
11/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
12/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
13/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
14/2/2011	30.52	20.67	0.01	5.82	15.17	2.98
15/2/2011	19.93	13.33	0.01	3.89	7.63	2.46
16/2/2011	29.88	20.00	0.01	5.93	12.30	4.40
17/2/2011	0.00	0.00	0.00	0.00	0.00	0.00
18/2/2011	10.44	6.67	0.00	1.98	3.90	1.69
19/2/2011	30.69	20.00	0.01	5.88	14.60	4.44
20/2/2011	10.51	6.67	0.00	1.98	4.97	1.61
21/2/2011	31.14	20.00	0.01	5.86	13.00	4.28
22/2/2011	31.25	20.67	0.01	5.87	14.37	4.18

Sample 1 : Data Mining results



1.3 Phase 3: Data Pre-processing

Data pre-processing comprises of 3 activities which is data cleaning, data integration, and data reduction.

- **Cleaning of the selected data from noisy data**

Data pre-processing includes activities that need to be done before extracting the pattern of data. This includes data cleaning, data integration, and data reduction. After selected the suitable data for the project, the data need to be cleaned first to remove any outlier and dirty data such as missing value or incomplete data.

To cleaned the data, any unnecessary or unused data need to be removed. As for this project, the data has some incomplete values. This data need to be removed by ignoring the tuple. Since the in one tuple has more than fifty percent missing, then the appropriate way to clean it is by ignoring it. Figure 4.5 shows the result of data after it has been cleaned. The data contain incomplete value has been removed because it can affect the result of the data.

date	temp	cond	tds	pH	turb	odo
2/2/2014	24.40	35.00	0.00	6.52	73.10	6.43
3/2/2014	24.49	36.00	0.00	6.47	63.70	6.30
3/2/2014	24.43	37.00	0.00	6.42	54.10	6.05
3/2/2014	24.55	39.00	0.00	6.36	36.30	5.74
4/2/2014	24.80	40.00	0.00	6.36	27.80	5.65
4/2/2014	24.69	40.00	0.00	6.34	16.50	5.52
4/2/2014	24.62	38.00	0.00	6.04	50.00	3.88

Figure 4.5 Result of cleaned data

- **Integrate and combine cleaned data into one dataset**

Data integration is the combination of different table into one table. This involved data from table water level and rainfall. Figure 4.6 shows the result of data integration. Water level data and rainfall data has been combine into one table together with water quality data. Furthermore, Figure 4.7 shows the combination of data from the year 2011 until 2014.

Sample 1 : Data Mining results

date	temp	cond	TDS	pH	turbidity	ODO	wL	RF
1/1/2012	26.08	25.67	0.02	3.72	1003.40	0.19	0.00	0.00
2/1/2012	26.96	26.33	0.02	3.70	918.50	0.24	0.00	0.00
3/1/2012	26.80	26.00	0.02	3.56	909.07	0.45	0.00	0.00
4/1/2012	26.62	26.33	0.02	3.66	871.73	0.28	0.00	0.00
5/1/2012	26.56	28.33	0.02	3.69	846.67	0.21	0.00	0.00
6/1/2012	26.68	30.00	0.02	3.66	781.80	0.10	0.00	0.00
7/1/2012	26.66	32.67	0.02	3.68	698.37	0.13	0.00	0.00
14/1/2012	25.53	29.00	0.02	3.73	1091.50	0.04	0.00	0.00
15/1/2012	24.53	35.67	0.02	3.41	1063.93	0.27	0.00	0.00
16/1/2012	24.47	33.00	0.02	3.43	1018.13	0.04	0.00	0.00
17/1/2012	25.12	36.50	0.02	3.67	950.60	0.04	0.00	0.00
27/1/2012	27.20	81.00	0.05	3.72	41.33	5.79	0.00	0.00
28/1/2012	27.00	81.33	0.05	3.50	38.80	1.06	0.00	0.00
29/1/2012	26.67	81.33	0.05	3.37	34.70	0.90	0.00	0.00
30/1/2012	26.61	81.67	0.05	3.35	32.60	0.77	0.00	0.00
31/1/2012	27.00	83.00	0.05	3.37	30.20	0.96	0.00	0.00
1/2/2012	27.18	86.00	0.05	3.42	34.03	0.87	0.00	0.00
2/2/2012	27.47	89.00	0.06	3.39	32.40	1.05	0.00	0.00

Data from water level and rainfall has been integrated into table of water quality

Figure 4.6 Result of data integration

date	temp	cond	tds	pH	turb	odo	wl	rf
2011-12-23	30.40	55.00	0.00	4.97	9.73	3.26	0	358.8297
2011-12-24	30.21	53.67	0.00	4.98	17.93	3.24	0	395.9263
2011-12-25	30.22	53.50	0.00	5.00	10.30	3.23	0	397.384
2012-01-01	26.08	25.67	0.02	3.72	1003.40	0.19	0.00	0.00
2012-01-02	26.96	26.33	0.02	3.70	918.50	0.24	0.00	0.00
2012-01-03	26.80	26.00	0.02	3.56	909.07	0.45	0.00	0.00
2013-03-20	29.94	69.33	0.04	-9.99	4.10	1.76	11.97	17.40
2013-03-21	30.13	67.67	0.04	-9.99	0.40	1.73	11.94	17.40
2013-03-22	30.36	68.00	0.04	-9.99	1.23	1.86	11.91	17.40
2013-03-23	30.33	67.00	0.04	-9.99	1.40	1.88	11.89	17.40
2014-02-02	24.40	35.00	0.00	6.52	73.10	6.43	11.98	0.00
2014-02-03	24.49	37.33	0.00	6.42	51.37	6.03	11.98	0.00
2014-02-04	24.70	39.33	0.00	6.25	31.43	5.02	12.26	0.00

Figure 4.7 Result of combination data

- **Data reduction**

Data reduction is the process where the data has been reduced in smaller amount compared to the original data. This is to increase the efficiency of analyzing the data. When there is large amount of data, it can cause time consuming to analyse the data.

	A	B	C	D	E	F	G
321	17/12/2014	8.86	15.33	0.01	1.32	16.53	1.67
322	18/12/2014	8.79	15.00	0.01	1.32	15.73	1.67
323	19/12/2014	17.45	29.00	0.02	2.57	29.93	4.00
324	20/12/2014	8.70	16.67	0.01	1.31	15.60	1.67
325	21/12/2014	17.37	38.33	0.02	2.66	29.50	2.67
326	22/12/2014	0.00	0.00	0.00	0.00	0.00	0.00
327	23/12/2014	16.61	21.33	0.01	2.37	24.90	4.00
328	24/12/2014	8.43	10.33	0.01	1.18	12.40	1.67
329	25/12/2014	25.37	30.67	0.02	3.54	33.67	4.67
330	26/12/2014	25.26	33.00	0.02	3.59	24.90	3.33
331	27/12/2014	8.43	10.00	0.01	1.21	5.83	1.33
332	28/12/2014	25.33	34.67	0.02	3.61	49.93	4.67
333	29/12/2014	25.46	30.67	0.02	3.65	22.27	5.67
334	30/12/2014	25.44	26.67	0.02	3.59	12.80	2.00
335	31/12/2014	17.00	16.67	0.01	2.38	12.23	1.67

The data has been reduced to 335 data form 1003 data

Figure 4.8 Result of data reduction



Sample 1 : Data Mining results

1.4 Phase 4: Data Transformation

Data transformation is the process of transforming the data into suitable format to ease the process of reading the data.

- **Transformation the data that already being integrated into one table**

Figure 4.12 shows the result of data transformation. The name of the attribute has been transform in readable format and the value of the data also has been transform into two decimal places and has been standardize for all data.

date	temp	cond	TDS	pH	turb	ODO	wL
14/2/2011	30.52	20.67	0.01	5.82	15.17	2.98	12.84
15/2/2011	19.93	13.33	0.01	3.89	7.63	2.46	12.74
16/2/2011	29.88	20.00	0.01	5.93	12.30	4.40	12.65
17/2/2011	0.00	0.00	0.00	0.00	0.00	0.00	12.56
18/2/2011	10.44	6.67	0.00	1.98	3.90	1.69	12.50
19/2/2011	30.69	20.00	0.01	5.88	14.60	4.44	12.42
20/2/2011	10.51	6.67	0.00	1.98	4.97	1.61	12.35
21/2/2011	31.14	20.00	0.01	5.86	13.00	4.28	12.28
22/2/2011	31.25	20.67	0.01	5.87	14.37	4.18	12.22
23/2/2011	31.12	21.00	0.01	5.86	12.97	4.07	12.17
24/2/2011	30.43	20.33	0.01	5.92	13.87	4.08	12.12
25/2/2011	30.18	21.33	0.01	5.96	16.33	4.11	12.08
26/2/2011	30.20	20.67	0.01	5.93	13.73	3.97	12.04
27/2/2011	30.38	21.33	0.01	6.15	15.87	4.01	12.00
28/2/2011	30.32	21.00	0.01	6.64	14.53	4.37	11.97
1/3/2011	30.13	21.00	0.01	7.15	12.70	4.41	11.93
2/3/2011	29.98	21.00	0.01	7.79	12.87	4.22	11.89
3/3/2011	29.60	20.33	0.01	7.66	17.37	4.23	11.86
4/3/2011	30.14	21.00	0.01	7.96	14.40	4.51	11.84
5/3/2011	30.66	21.00	0.01	7.74	14.43	4.35	11.81

Figure 4.12 Result of data transformation

#	Name	Type	Collation	Attributes	Null	Default
1	parameterID	int(11)			No	*None*
2	date	date			No	*None*
3	temp	decimal(10,2)			No	*None*
4	cond	decimal(10,2)			No	*None*
5	tds	decimal(10,2)			No	*None*
6	pH	decimal(10,2)			No	*None*
7	turb	decimal(10,2)			No	*None*
8	odo	decimal(10,2)			No	*None*
9	wl	decimal(10,2)			No	*None*



Sample 1 : Data Mining results

1.5 Phase 5: Naïve Bayes Analysis

In phase 5, the process of analyze the data using the suitable data mining process. There are a few research and studies that has been done to find the suitable data mining technique which was discussed in chapter 2, Literature review of the project. Naïve Bayes technique is chosen because it is the most suitable data mining technique. Based on the result in chapter 2, it shows that Naïve Bayes technique has many benefit compare to others technique. This technique will analyze the data and display the result of the data in the form of probability.

- **Identify the prior probability using Naïve Bayes technique**

Table 4.1 Result of Prior Probability

PRIOR PROBABILITY			
	COUNTS	**FORMULA**	**PROBABILITY**
P(class = I)	1169	1169/2128	0.54
P(class = II)	375	375/2128	0.18
P(class = III)	266	266/2128	0.13
P(class = IV)	157	157/2128	0.07
P(class = V)	161	161/2128	0.08

Table 4.1 shows the result of prior probability that will be used to calculate the conditional probability. The prior probability calculated are including the class label which is class I, class II, class III, class IV, and class V. This five class



Sample 1 : Data Mining results

For example, given the value of tuple x is temp = A, cond = A, tds = A, pH = C, turb = B, odo = D, and wL = A. Based on Table 3.4, in order to know the particular value will belong to which class level, we need to calculate each of the attribute given.

Table 4.3 Selected Attribute

CONDITIONAL PROBABILITY			
CLASS		**FORMULA**	**PROBABILITY**
I	P(temp = A | I)	304/1169 = 305/1176	0.2594
	P(cond = A | I)	304/1169 = 305/1176	0.2594
	P(tds =A | I)	304/1169 = 305/1176	0.2594
	P(pH = C | I)	0/1169 = 1/1176	0.0009
	P(turb = B | I)	0/1169 = 1/1176	0.0009
	P(odo = D | I)	0/1169 = 1/1176	0.0009
	P(wL = A | I)	174/1169 = 175/1176	0.1488
II	P(temp = A | II)	0/375 = 1/382	0.0026
	P(cond = A | II)	0/375 = 1/382	0.0026
	P(tds =A | II)	0/375 = 1/382	0.0026
	P(pH = C | II)	0/375 = 1/382	0.0026
	P(turb = B | II)	304/375 = 305/382	0.7984
	P(odo = D | II)	0/375 = 1/382	0.0026
	P(wL = A | II)	0/375 = 1/382	0.0026
III	P(temp = A | III)	0/266 = 1/373	0.0027
	P(cond = A | III)	0/266 = 1/373	0.0027
	P(tds =A | III)	0/266 = 1/373	0.0027
	P(pH = C | III)	128/266 = 129/373	0.3458
	P(turb = B | III)	0/266 = 1/373	0.0027
	P(odo = D | III)	0/266 = 1/373	0.0027
	P(wL = A | III)	0/266 = 1/373	0.0027
IV	P(temp = A | IV)	0/157 = 1/164	0.0061
	P(cond = A | IV)	0/157 = 1/164	0.0061
	P(tds =A | IV)	0/157 = 1/164	0.0061
	P(pH = C | IV)	0/157 = 1/164	0.0061
	P(turb = B | IV)	0/157 = 1/164	0.0061
	P(odo = D | IV)	72/157 = 73/164	0.4451
	P(wL = A | IV)	0/157 = 1/164	0.0061



Sample 1 : Data Mining results

1.6 Phase 6: Pattern evaluation

In this phase, the pattern of data will be analyzed in order to get the result of the water quality. This is to know the water will belong to which class level. The pattern of data is according to the probability that has been calculated in phase 5. This pattern need to be validated and evaluate first before embedding it into the system.

- **Evaluate and validate rules extracted from the data**

Pattern of evaluation is the result get after calculating the prior probability, conditional probability, and posterior probability. This section will give example of result of each pattern generated for class I, class II, class III, class IV, and class V.

IF P(temp=A | class=I) = 0.2594
P(cond=A | class=I) = 0.2594
P(tds=A | class=I) = 0.2594
P(pH=A | class=I) = 0.0706
P(turb=A | class=I) = 0.0009
P(odo=A | class=I) = 0.0009
P(wL=A | class=I) = 0.1488
THEN P(x|I) = 0.0000000001485

According to the first sample above, it shows that the if all of the attribute fall under range A, the attribute will belong to class I. All of the value for the probability was taken from conditional probability value that has been calculated and the value need to be multiply by each other to get the value of 0.0000000001485.



Sample 1 : Data Mining results

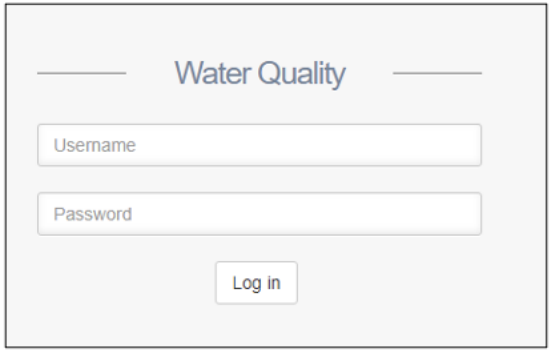


Figure 4.14 Login Page

Figure 4.14 shows the login page of the system. In this page, user is required to insert their username and password to login to the system. This is because, this can allow only authorize user that can login to the system and to protect the data from being modified by other unauthorized user.

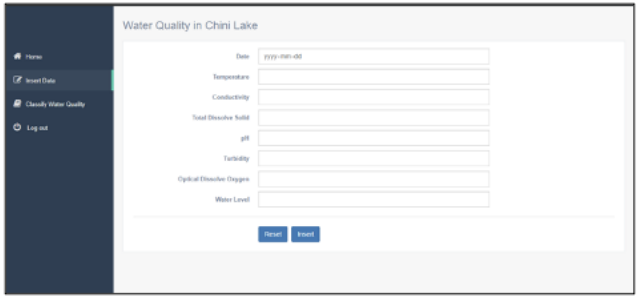


Figure 4.15 Insert data page

Figure 4.15 shows the insert data page. For this page, user can insert the water quality data that they took to be stored into the database. User need to insert the date they took the data, and all other 7 attribute in order for the data to be always updated and can hence can classify those data will belong to which class level.



Sample 1 : Data Mining results

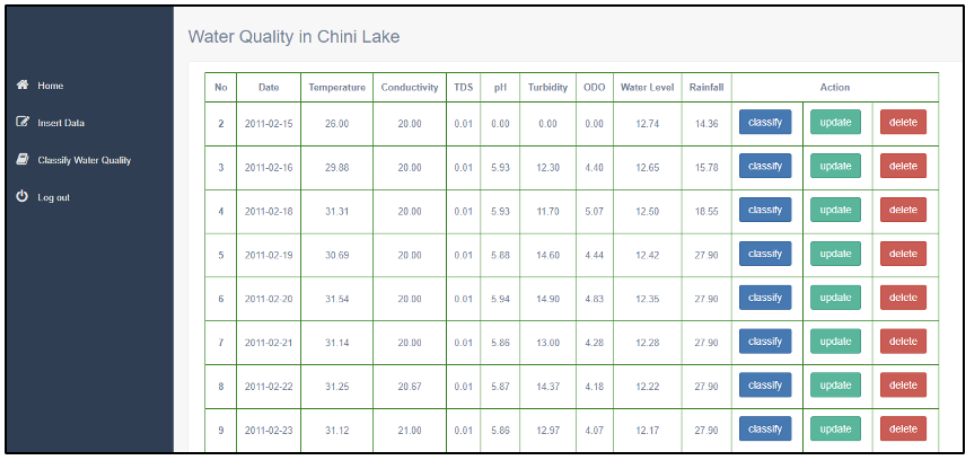


Figure 4.16 Water quality classification page

Figure 4.16 shows the water quality classification page. In this page, user can see the overall data and can perform action to the data. User can either classify the data, update the data, or can delete the data. When user wanted to know in that particular day of data will fall under which class, user can click classify button. After clicking the button, user will be directed to the classification page. In this page, it will show the result of water quality after calculating the data and the probability for each class level. Next, for updated function, user need to click the update button to update any particular value of data. other than that, when user wanted to delete the data, user can simply delete the data by click the delete button.



Sample 2 : Rule-based Results



4.2.1 Rule Based Design

The first activity in this phase was generating and analysing rules. Rules were generated to be used as the criteria and constraints for student's selection in the system. The Figure 4.4 shows the sample rules that has been generated and analysed. The full list of rules can be found in Appendix B of this report. The rules are the zakat criteria which the staff store by himself. The rules are consisting of six attributes, which are the parent's salary, family members, students loan, students spending, transportation, student's current examination result the reason for students to apply zakat. The value for every attribute is either 1, 2, 3 or 4. The values 1,2,3 and 4 were used to simplify the rates of every questions. This can be shown in table 6.

Table 6 Rate Value for Each Criteria

Zakat Criteria	Value for each Zakat Criteria
Salary (RM)	1) 800 – 1200 2) 1201 – 1500 3) 1501 - 2000 4) 2000 dan ke atas
Family members (excluded married siblings)	1) 1 – 3 2) 4 - 7 3) 8 - 11 4) 12 dan ke atas
Students Loan	1) PTPTN 2) JPA 3) Biasiswa 4) Bantuan Lain
Students Spending (RM)	1) 100 – 150

Sample 2 : Rule-based Results

Table 6 (cont.)

	2) 151 - 200 3) 201 - 300 4) 300 dan ke atas
Students Transportation	1) Kereta 2) Motorsikal 3) Basikal 4) Jalan kaki
Student Examination Result (CGPA)	1) 1.50 – 2.50 2) 2.51 – 2.99 3) 3.00 – 3.50 4) 3.51 – 4.00
Why the students apply zakat?	1) Sebagai duit tambahan untuk menampung perbelanjaan 2) Sangat memerlukan zakat 3) Ingin mencuba tahap kelayakan zakat 4) Semua di atas

However, there are sufficient descriptions displayed on the webpage for every rate to ensure that the students rate themselves accordingly. The staff is the one who set the rules for this system.

Table 7 Example of rules inserted by staffs

No.	Rules Generated
1	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 1 THEN appStatus= selected
2	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 2 THEN appStatus= selected
3	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 3 THEN appStatus= selected
4	IF salary = 1 AND familyMember = 1 AND studLoan = 1 AND studSpending = 1 AND studTransport = 1 AND studResult = 1 AND studReason = 4 THEN appStatus= selected



Sample 2 : Rule-based Results

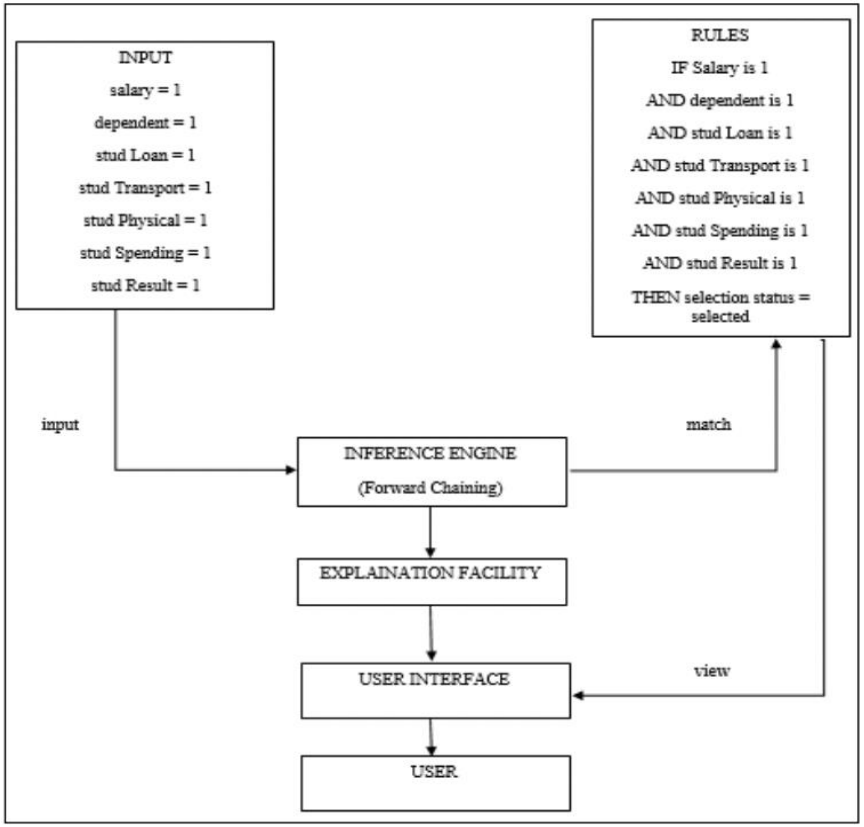


Figure 4.4 Rule Based Architecture for Online Zakat Management

According to the Figure 4.4, the facts for Online Zakat Management include salary, family members, student loan, student spending, student transport, student result, and student reason. These facts are entered by the user which is the students through the application form. After the students have entered all the facts, the facts will be processed by the inference engine of the system to be matched with the rules that have been set in the database.



Sample 2 : Rule-based Results



Figure 4.5 Welcome Page of Online Zakat Management System

From the welcome page, the user can be redirected to two other pages which are 'Login Staf' and 'Login Pelajar'. For staff and student that already have an account, they can click the 'Login' button to access the system. However, for students and staffs who do not have an account, they can register an account by clicking the 'Register Account' button.

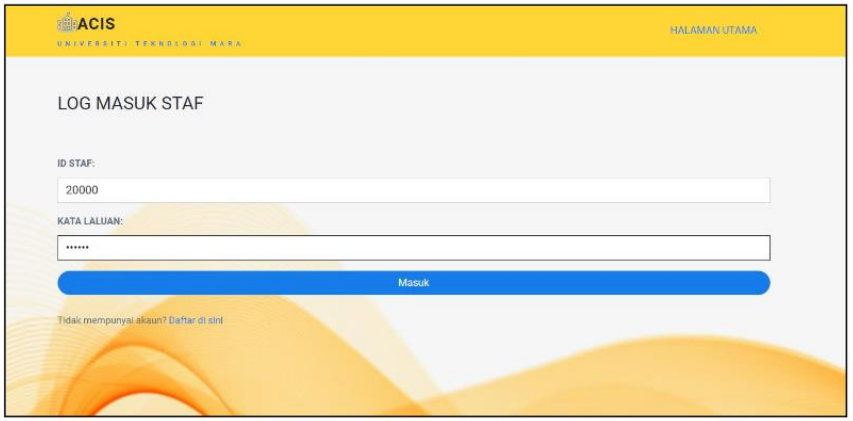


Figure 4.6 Staff login page



Sample 2 : Rule-based Results

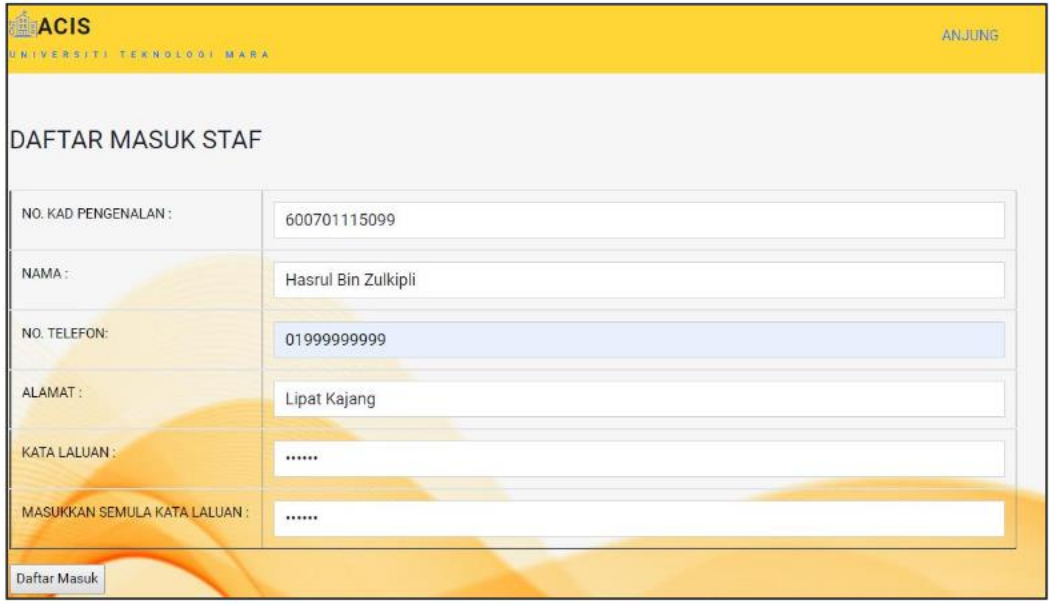


Figure 4.7 Staff registration form

Figure 4.7 shows the registration page for staffs who do not have an account. In the registration form, staff must enter 6 personal details such as name, phone number, address, identification card, password and retype password. Next, staff must click the 'Submit' button to register their account. Staff also can click the 'Cancel Registration' button to clear all fields in the form. After staff clicked the 'Submit' button, staff will be redirect to the staff login page as shown in Figure 4.8. The following page after staff had logged in is the homepage where it shows the details staff entered earlier when registering their account.



Sample 2 : Rule-based Results



Figure 4.8 Student login page

Figure 4.8 shows the login interface for Student. Similar to the staff login page, there are also two fields that the student must input which are the student ID and the password. Both inputs must be entered correctly to ensure a successful login by the student. Other than that, there are two links in this page which are the 'Register Here' and 'Login here'. The link 'Register Here' will redirect student to the account registration page whereas the 'Login here' link will redirect the user to staff login page.

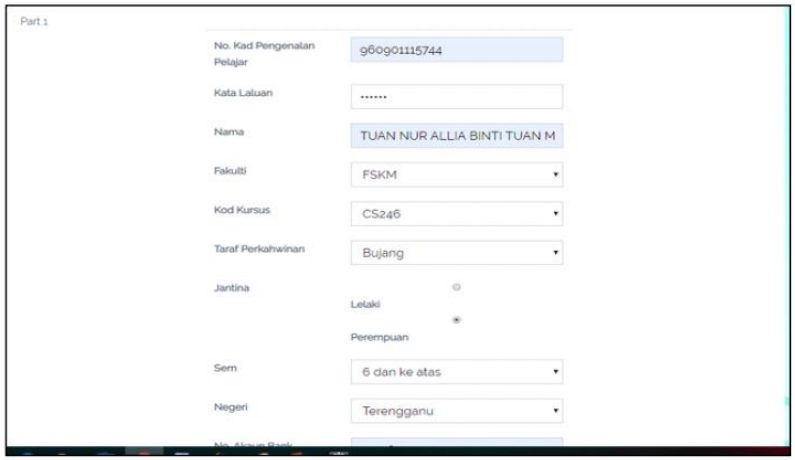


Figure 4.9 Student registration form



Sample 2 : Rule-based Results



Figure 4.12 shows the homepage of students. At the top of the page, there is a navigation bar that contains few buttons. Those buttons are 'Anjung Pelajar', 'Permohonan Zakat', 'Semakan Keputusan Zakat', 'Akaun Saya, and 'Log Keluar' button. These buttons will redirect student to different pages. To apply the zakat, the student need to click on the 'Permohonan Zakat'.



Figure 4.15 Student Application Form



Figure 4.16 Student Application Form

Figure 4.15 and 4.16 shows the application form that must be fill by the students to apply the zakat. There are 13 personal details that had been provided which are name, identification card, faculty, course code, gender,

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Sample 2 : Rule-based Results

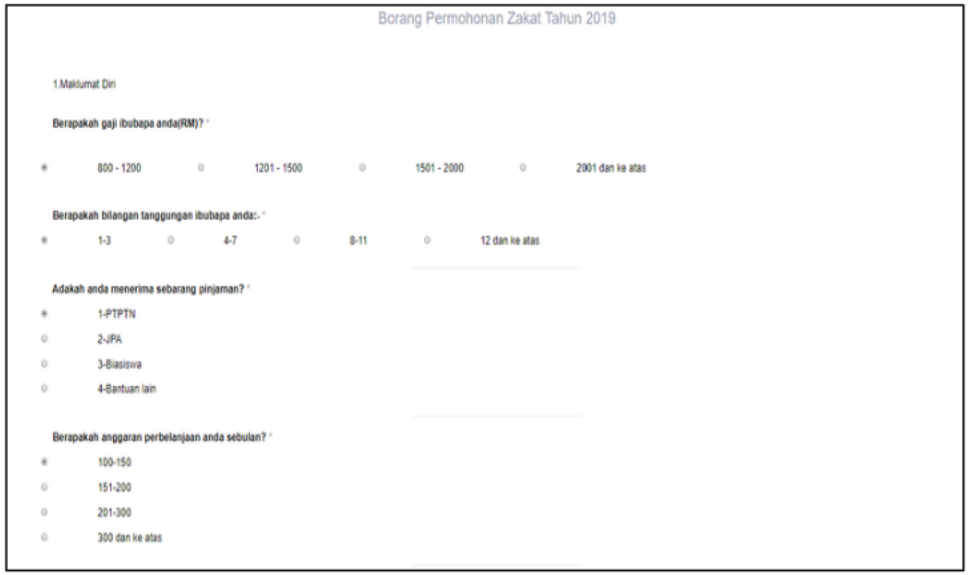


Figure 4.17 Student Criteria Form

Figure 4.17 shows the form that must be filled by the students. The students need to click 'Student Criteria Form' at the upper part of interface. This page contains all the questions that students need to answer in order to apply the zakat. The first question is about parents' salary. The second questions is the amount of family members. Third question is students loan or scholarship.



Sample 2 : Rule-based Results



SENARAI NAMA PENERIMA ZAKAT 2019

Tahniah kepada pelajar yang layak menerima zakat. Kepada pelajar yang tidak mempunyai nama dalam senarai, nama anda masih berada di senarai menunggu. Anda boleh klik pada nama anda untuk mengetahui kelayakan anda.

No.	ID Pelajar	Nama Pelajar
1.	2016655642	Aimi Nazirah Binti Zakaria
2.	2016655698	Siti Aisyah Binti Zainal Abidin
3.	2016655706	Siti Nurhaliza Binti Shah Azam
4.	2016655688	Michael Anggun
5.	2016655642	Sabarina Saad
6.	2016655698	Zahirah Manaf
7.	2016655706	Nur Eliza Shukor
8.	2016655688	Amirul Hakim Jaafar

Figure 4.19 View Selected Student Page

The Figure 4.19 shows the interface when student clicks the 'Semakan Pelajar Terpilih' button. In this page, the names of selected students will be displayed along with their respective student id. If a student finds their own name in the selected student list, their name will appear in blue fonts. They can click on their name to view the explanation facility.

SENARAI NAMA MENUNGGU ZAKAT 2019

Nama anda masih berada di senarai menunggu. Anda boleh klik pada nama anda untuk mengetahui kelayakan anda.

No.	ID Pelajar	Nama Pelajar
1.	2016655642	Aimi Nazirah Binti Zakaria
2.	2016655698	Siti Aisyah Binti Zainal Abidin
3.	2016655706	Siti Nurhaliza Binti Shah Azam
4.	2016655688	Tuan Nur Allia Binti Tuan Mohd Zamli

Figure 4.20 Waiting list

Sample 2 : Rule-based Results



suhailayani@uitm.ed

4.3.1 Example of System Result

In this section, two results of the system will be presented to show the differences when the input of the students matched with the rules as well as when the input of the students unmatched with the rules.

a) Student A.

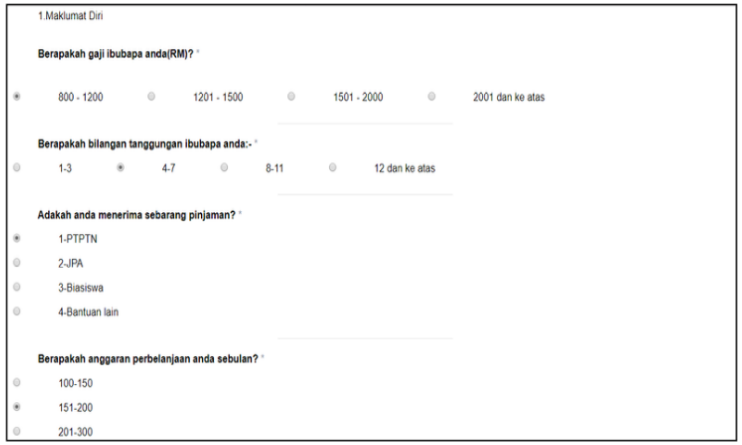


Figure 4.23 Input entered by student A

Figure 4.23 shows four out of seven inputs entered by Student A. Student A choose salary and dependents in rating 2 while for student loan and student spending in rating 1.



Figure 4.24 Input entered by student A (cont.).

Sample 2 : Rule-based Results



Figure 4.24 shows the remaining inputs from Student A. Student A chose rating 3 for examination result and reason for applying zakat. Student A also ticked the checkbox to agree to the term and condition.



SENARAI NAMA PENERIMA ZAKAT 2019

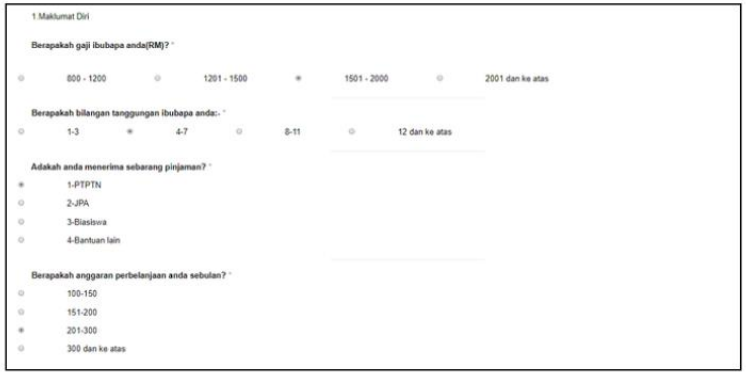
Tahniah kepada pelajar yang layak menerima zakat. Kepada pelajar yang tidak mempunyai nama dalam senarai, nama anda masih berada di senarai menunggu. Anda boleh klik pada nama anda untuk mengetahui kelayakan anda.

No.	ID Pelajar	Nama Pelajar
1.	2016655642	Aimi Nazirah Binti Zakaria
2.	2016655698	Siti Aisyah Binti Zainal Abidin
3.	2016655706	Siti Nurhaliza Binti Shah Azam
4.	2016655688	Michael Anggun
5.	2016655642	Sabarina Saad
7.	2016655706	Nur Eliza Shukor
8.	2016655688	Amirul Hakim Jaafar

Figure 4.25 Result of system for Student A.

Figure 4.25 shows the result of system for Student A. Student A name appeared at the View Selected Player page. Student A is selected because all the inputs entered by Student A matched with one of the rules in the database. The name of Student A is in blue font to show that Student A can click on own name to read the explanation facility.

b) Player B.



1.Maklumat Diri

Berapakah gaji ibubapa anda(RM)? *

800 - 1200 | 1201 - 1500 | 1501 - 2000 | 2001 dan ke atas

Berapakah bilangan tanggungan ibubapa anda:- *

1-3 | 4-7 | 8-11 | 12 dan ke atas

Adakah anda menerima sebarang pinjaman? *

- 1-PTPTN
- 2-JPA
- 3-Biasiswa
- 4-Bantuan lain

Berapakah anggaran perbelanjaan anda sebulan? *

- 100-150
- 151-200
- 201-300
- 300 dan ke atas

Figure 4.26 Input entered by student B.

Sample 2 : Rule-based Results

To be conclude, based on the examples shown previously, there are two possible outcomes from this system. The outcomes depend on the inputs entered by the students. A student can whether be selected or not selected. If the inputs of student matched with any rules in the database, student will be selected, and the name of student will be displayed in View Selected Student page. In contrast, if the inputs of student do not match with any rules in the database, the student will not be selected, and the name of student will be displayed in the Waiting List page.

Table 9 Summary for Student A and Student B system results.

	Inputs Entered	**Rules (constraint) matched? (%)**	**Result**
Student A	salary = 2 dependent = 2 stud Loan = 1 stud Transport = 1 stud Physical = 1 stud Spending = 1 stud Result = 1	100%	Pass to get the zakat
Student B	salary = 3 dependent = 1 stud Loan = 1 stud Transport = 2 stud Physical = 1 stud Spending = 1 stud Result = 1	0%	Waiting list

Table 10 summarized the inputs entered by Student A and B, as well as whether their inputs matched with any rules in the database and the id for the rule matched.



Sample 2 : Rule-based Results

Table 10 Current process vs. Online Zakat

Current Process	Online Zakat Management System
Apply zakat through downloading the form.	Application can be done using Online Zakat System.
Selections being done manually and affected by human biasness.	Selections being done by the inference engine of the system and not affected by human biasness.
List of selected students display at ACIS noticeboard.	List of selected students can be viewed through Online Zakat Management System.

4.4 Chapter Summary

This chapter consists of all the deliverables, findings and results for each phase of SDLC that this project undergoes. All the results were explained in detailed as well as how those results achieved the project objectives were explained too. There are also screenshots of the system interface presented in this chapter.



To – do for this week

- Complete drafting your chapter 4
- Please submit and discuss your current Chapter 4 with your supervisor.
- You have 3 more weeks before progress presentation (starts at week 8 until week 14)**, please ensure you have system progress to present.**

